Eswar Adapa

Data Scientist

GitHub | LinkedIn

(+91) 6309925636
Mumbai, India
eswaradapa90@gmail.com

Professional Summary

- Data Scientist with experience in **Machine Learning, Large-Scale Data Engineering, Graph Systems, and GenAI**. Focuses on solving complex data problems and delivering **scalable, production-grade systems**.
- Led multiple high-impact initiatives, scaling the **POI ingestion pipeline from 23M to 220M records**, reducing geospatial processing time from **1 week to 2.5 hours**, and developing a multi-layer **POI Knowledge Graph** that improved downstream model performance by **15–20%**. Migrated the GenAI stack to a **LangGraph multi-agent system**, increasing throughput by **10×** and supporting **5000+ concurrent users**. Received **Best Employee (Feb 2024 – May 2025)**.
- Demonstrates strong **ownership**, remains **honest**, and consistently exhibits a **hardworking** approach in all responsibilities.

Skills

Languages	Python, MATLAB, Embedded C
ML & AI	Machine Learning, Deep Learning, NLP, Generative AI, Transformers (BERT, DistilBERT), Recommender Systems, OCR (Tesseract), Image Processing
LLM Systems	LangChain, LangGraph, RAG, Multi-Agent Systems, Prompt Engineering, Model Routing, Guardrails
Data & Backend	PySpark, pandas, FastAPI, REST APIs, Pydantic, WebSockets, Distributed Data Processing
Databases	SQL, MongoDB, Redis, Qdrant, FAISS
Graph & Spatial	Neo4j, GDS, Apache Sedona, H3 Indexing
Cloud & DevOps	AWS (EC2, S3, EKS), Docker, Kubernetes, Jenkins

Experience

Data Scientist, Infinite Analytics — **Feb 2024 – Present**

- **POI Ingestion and Data Scaling**
  - Unified POI crawling and ingestion pipelines using RabbitMQ, Redis, and MongoDB, scaling coverage from **23M POIs across 2 countries** to **220M POIs across 18 countries**.
  - Built and deployed a **validation backend** that enabled large-scale ground-truth collection for POI quality checks.
  - Improved **POI-type mapping accuracy from 83% to 87.3%** by introducing a **Name–Tag Mapper** and leading large-scale **model training and architecture optimization**, evolving from a **hierarchical LightGBM cascade (51 models)** to a unified **DistilBERT transformer** with **8-head self-attention, multi-feature fusion, class-imbalance reweighting, and hard-negative mining**; trained on **2.5M POIs**, achieving **96.7% accuracy** and **0.005s inference latency**.
  - Enhanced geospatial accuracy using **plus-code mappings** and **triangular normalization**.
  - Accelerated Point-to-Polygon mapping with Apache Sedona, reducing runtime from **1 week to 2.5 hours**; introduced a **partition-aware pre-join strategy** (global split into 12 population-weighted regions with internal heuristics) and **salting** to optimize join performance and scalability.
- **Sherlock AI: Multi-Agent Intelligence & Decision Platform**
  - Migrated Sherlock AI from monolithic LangChain AgentExecutor (20+ tools, 300-line prompt, SQLite, limited memory) to a **LangGraph multi-agent StateGraph** (SearchAssistant → 6 specialists), with **MongoDB persistent state** (TTL, checkpointing) and **Langfuse tracing**; achieved **10× throughput (100 req/s)**, **70% prompt reduction**, **30% lower token cost**, **60% faster responses (3–5s → 1–2s)**, supporting **5000+ concurrent users**.
  - Built a **3-phase onboarding agent** (Research & Q&A → Audience Strategy → Reports) using web search, **Pydantic validation**, and strict evidence labeling; added **checkpoint-based recovery**, WebSocket interface, and persisted outputs (JSON, Excel, Word) in versioned runs.
  - Implemented **plan-aware guardrails**: action classification (SQL / insight / multi-step), auto-executed SQL, human-validated insights, and **Qdrant-based semantic schema retrieval** for accurate query generation—reducing hallucinations and separating reasoning vs execution.
  - Designed **multi-layer RAG**: Qdrant (schema), FAISS (fuzzy brand match), ChromaDB (POI resolution); built a **LLM factory** (OpenAI, Anthropic, Cohere, Google, HuggingFace) with config-driven routing and failover.
  - Improved reliability with **JWT auth (RBAC)**, MongoDB TTL sessions, and observability (Loguru, Datadog, New Relic); deployed via Docker, Jenkins (SonarQube), and Kubernetes (AKS).
- **POI Knowledge Graph (Neo4j + GDS + Python)**
  - Improved an existing **POI Knowledge Graph** integrating multiple datasets (POI, visitation, behaviors, WBI, reviews, touchpoints) to support more efficient cross-dataset querying.
  - Implemented graph embedding pipelines using **Node2Vec, FastRP, HashGNN, and GraphSAGE** to generate multi-modal embeddings combining attribute and structural features.

- Optimized graph pipelines handling large-scale nodes and relationships to improve embedding generation and processing performance.
- Built similarity relationships (`SIMILAR_POI_TYPES`, `SIMILAR_H3INDEX`, `SIMILAR_PERSON`) using **Neo4j GDS KNN**, supporting downstream **recommendation and segmentation models**.

Research Intern, DRDO — **May 2023 – Aug 2023**

- **sEMG Signal Analysis And Gait Modeling**
  - Improved preprocessing and symbolization of sEMG data, yielding a 20% increase in data quality
  - Trained deep learning models on IMU and sEMG walking data to classify 7 gait phases with 92% accuracy
  - Applied Symbolic Transfer Entropy (sTE) to study muscle interactions across gait phases, identifying a link between walking speed and a 25% rise in inter-muscle information transfer

Publications

Fatigue Classification and Onset Estimation Using sEMG Signals During Strength Training — **Aug 2022 – Apr 2023**

APSIPA'23 – IEEE

- Studied fatigue classification and onset estimation using sEMG signals during strength training; the work was accepted at the APSIPA'23 conference in Taiwan.
- Reached 86% classification accuracy and coined a Machine Learning model with Majority Windowing, achieving a 12% estimation error.

Education

Indian Institute of Information Technology, Sricity — **Dec 2020 – Apr 2024**

B.Tech.(Honors), Electronics and Communication Engineering

- CGPA: 8.02

Achievements

Best Employee Award — **Feb 2024 – May 2025**

- Recognized with the **Best Employee Award** for notable contributions to Data Engineering, Knowledge Graph development, and GenAI integration across multiple projects.

Academic Grand Challenge on Climate Change, Nasscom and Telangana Government — **Feb 2023 – Mar 2023**

- Devised a reliable predictive framework for heatwave events and Air Quality Index in Tier-2 cities of Telangana, achieving better performance than existing baselines.
- Ranked among the top 40+ teams out of 7000+ participants in the nationwide competition.

Position of Responsibility

Internship Manager – BITS Pilani Collaboration — **Jun 2025 – Jul 2025**

- Served as **Scrum Master and Project Manager** for a cohort of 5 BITS Pilani interns during an 8-week program.
- Directed task allocation, sprint planning, and progress tracking to ensure timely project completion.
- Guided interns on technical and workflow practices, supporting effective collaboration in a fast-paced setting.

Secretary – IEEE, IIIT Sricity — **Mar 2023 – Apr 2024**

- Coordinated travel and lodging for 100+ attendees at the ESDC Conference, ensuring accurate and seamless arrangements.
- Managed conference logistics, contributing to an event rated **4.5/5** by participants.